

Recommended Assessment

Infrared Distance

Observing Infrared Distance Data

1. What do the raw voltage measurement from the Infrared Distance sensor tell you about how the sensor behaves?
2. What do the minimum and maximum values tell us about the range of the Infrared distance sensor?

Calibrating for Infrared Distance Sensor

3. How do the calibrated A and b values match the sampled data?
4. Does your graph look similar to the distance measuring graphs shown in the datasheet? What can you observe in the graph in terms of the object color used to measure distance to?

Testing Calibration

5. Using the calibrated A and b values, does the reported distance measurement match the calibration ruler? What other considerations might you need to take when using an infrared distance sensor?

Real Life Scenarios

6. You're designing a series of mobile robots which use infrared distance sensors to detect the presence of objects as obstacles. You notice there are random instances when the mobile robot enters an obstacle detection routine. This effect becomes worse when you go outdoors. What are some design considerations for the Infrared distance sensor which could be causing this issue?
7. You're using an infrared distance sensor to count objects passing through a conveyor belt. You've noticed that metallic objects are sometimes counted incorrectly. What are some design changes you can make to increase the reliability of the object counter?

Infrared Distance Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	